

Day 14: Advanced DAX Filter Modifiers

[DEF] Definition to Remember

Filter modifier functions (ALL, ALLSELECTED, REMOVEFILTERS, KEEPFILTERS, FILTER) work inside CALCULATE() to explicitly control which dimension filters are removed, preserved, or dynamically evaluated.

1. `ALL()` vs `REMOVEFILTERS()` -- Calculating % of Total

ALL() removes all filters from a table or column, returning all rows:

```
// Grand Total
Grand Total =
CALCULATE(
    [Total Sales],
    ALL(Product)
)

// Contribution %
Contribution % =
DIVIDE(
    [Total Sales],
    Grand Total
)
```

2. `ALLSELECTED()` -- Dynamic Visual Contribution %

ALLSELECTED() removes visual internal axis filters while preserving external user slicer selections:

```
Sales Contribution % =
DIVIDE(
    [Total Sales],
    CALCULATE(
        [Total Sales],
        ALLSELECTED(Product)
    )
)
```

3. The `FILTER()` Iterator Function

FILTER(Table, Condition) iterates through a table row-by-row to apply complex measure-based filtering:

```
Sales Filtered =
CALCULATE(
    [Total Sales],
    FILTER(
        Product,
        [Total Sales] > 0
    )
)
```

[MUST-WRITE] Must-Master Skills & Interview Points

1. Differentiate between ALL() (ignores everything) and ALLSELECTED() (respects user slicers).
2. Why REMOVEFILTERS() is the modern, semantic replacement for ALL() as a filter modifier.
3. Performance tip: Do NOT wrap simple column comparisons inside FILTER() unless filtering by a dynamic measure.

[RECALL] Quick Recall

ALL (Clear all filters) + ALLSELECTED (Clear axis but keep slicers) + FILTER (Table iterator for complex criteria)